



Universität Hamburg

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Universitätsklinikum
Hamburg-Eppendorf

Predicting Protein Crystallization Conditions

Data Acquisition, Preprocessing, Exploratory Data Analysis

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Maschinelles Lernen in der Bioinformatik

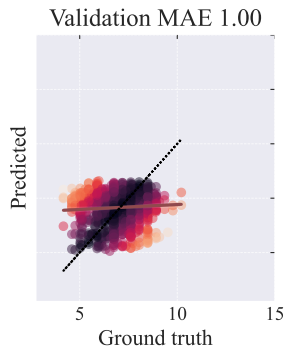
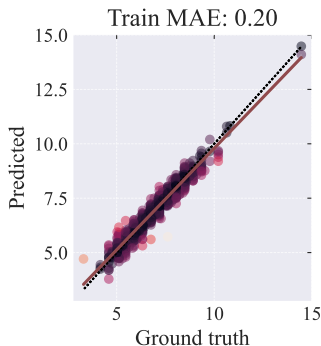
January 27, 2026

- ▶ protein based validation split instead of entry based
- ▶ build bi-partition tree to avoid data leakage from files with multiple proteins
- ▶ Explained Variance:
 - ▶ Train: 0.96
 - ▶ Validation: 0.25
- ▶ Pearson:
 - ▶ Train: 0.98
 - ▶ Validation: 0.5

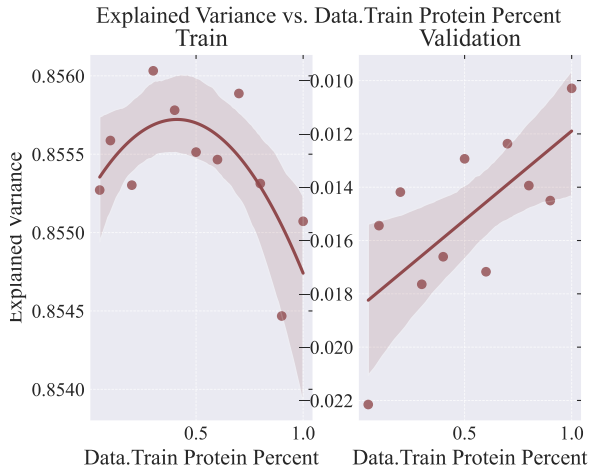
Ph Model Performance.

Naive:

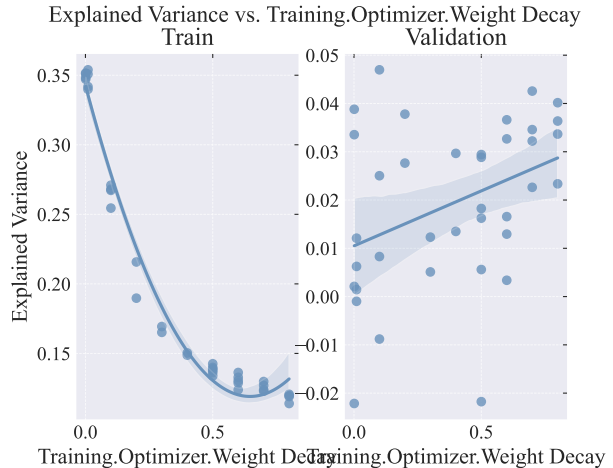
Train: 0.90, Val: 0.90



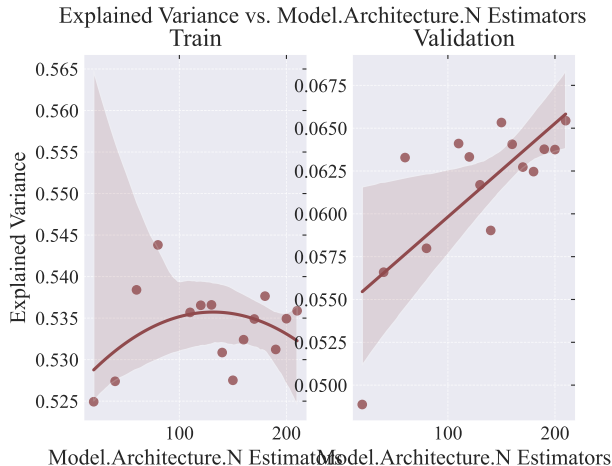
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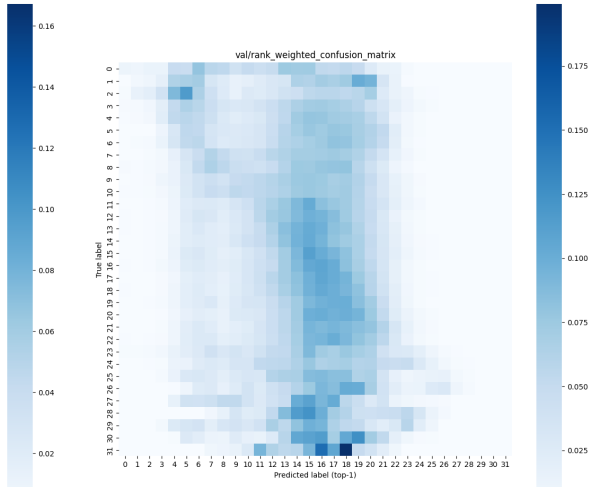
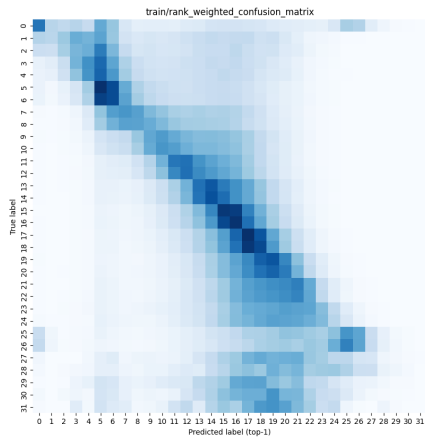
- ▶ How much data do I actually need?
- ▶ **Does penalizing help?**
- ▶ Does making the model smaller/larger help?



- ▶ **Problem setting:** We observe *positive* samples ($y = +1$) and *unlabeled* data. Negative labels ($y = -1$) are never directly observed. Bekker and Davis (2020)
- ▶ **Goal:** Learn a binary classifier $f(x)$ that separates positives from negatives despite missing negative labels.
- ▶ **PU risk (no negative labels required):**

$$R_{\text{PU}}(f) = \underbrace{\pi_+ \mathbb{E}_+[\ell(f(x), +1)]}_{\text{learn what "pos" looks like}} + \underbrace{\mathbb{E}_U[\ell(f(x), -1)]}_{\text{discourage predicting pos everywhere}} - \underbrace{\pi_+ \mathbb{E}_+[\ell(f(x), -1)]}_{\text{remove hidden-pos penalty}}$$

- ▶ **Intuition:** Treat unlabeled data as negative *on average*, then subtract the hidden positives.



Bekker, Jessa and Jesse Davis (Apr. 2020). "Learning from positive and unlabeled data: a survey". In: *Machine Learning* 109.4. arXiv:1811.04820 [cs], pp. 719–760. ISSN: 0885-6125, 1573-0565. DOI: 10.1007/s10994-020-05877-5. URL: <http://arxiv.org/abs/1811.04820> (visited on 01/27/2026).